12 A light wave of amplitude A is incident normally on a surface of area S. The intensity reaching the surface is I.

The amplitude of the light wave is increased to $2A$. The light is then focused on to a smaller area $\frac{1}{3}S$.

What is the intensity on this smaller area?

A $6I$ **B** $12I$ **C** $18I$ **D** $24I$

Ans: B
Intensity (power per unit area) is proportional to the square of amplitude: $I \propto A^2$

Effect of doubling the amplitude
Original intensity (power per unit area) = I, with amplitude A.
When amplitude is doubled to $2A$: $I_{new} = k(2A)^2 = 4kA^2 = 4I$

Effect of focusing onto smaller area
The same total power is now focused onto area $\frac{1}{3}S$:
New power per unit area $= {}^{4I \times S}\!/_{\frac{1}{3}S} = 4I \times 3 = 12I$